

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, C++, Java, SQL, Bash

LLM & Agentic Systems

LLM agents, tool use, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama, MCP

Retrieval / Search / RAG

RAG, RRF, cross-encoder reranking, MMR, knapsack token budgeting, Brave Search, web scraping, cell-level provenance, sentence-transformers

ML / Deep Learning

PyTorch, JAX, HuggingFace Transformers, NumPy, normalizing flows, diffusion, autodiff

Backend & Cloud

FastAPI, Node.js, Express, REST APIs, async services, Docker, AWS, GCP, Azure, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability, remediated **20+ VAPT findings**, and ran knowledge-transfer sessions that onboarded a second developer to on-call readiness in ~6 weeks.
- Joined a **GitHub Copilot** productivity pilot, routing tech-debt PRs, code migrations, and security-finding triage through AI-assisted workflows and maintaining a versioned `copilot-instructions.md` conventions file to keep AI-generated code aligned with team patterns.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product for context-aware replies (prompt design, response-quality heuristics) and shipped an open-source **LLM auto-commenting feature end-to-end with no ML mentor on the team**, delivering customer-facing AI independently while building **20+** Node.js + Fastify REST APIs over PostgreSQL.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables using typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an evaluation-improvement agent that proposes new queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an edge-native AI research assistant: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step tasks survive process restarts and tool failures, with conversation state persisted in D1.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.